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ASSIGNMENT 2

6.13, 6.19, 6.20 (English texthook)

Chương 2, bài 9 (Giáo trình Thiết kế Luận lý số 2)

6.13 (State minimization) Minimize the states for the FSM given in Figure P6.13, using:

(a) State partitioning
(b) An implication table

PRESENT STATE	NEXT STATE / OUTPUT		
	AB = 00	AB = 01	AB = 10
s_0	s_4 / 1	s_2 / 0	s_1 / 1
s_1	s_2 / 0	s_5 / 1	s_4 / 1
s_2	s_1 / 1	s_0 / 0	s_3 / 1
s_3	s_2 / 0	s_5 / 1	s_4 / 1
s_4	s_0 / 0	s_5 / 1	s_1 / 1
s_5	s_2 / 0	s_4 / 1	s_2 / 1

FIGURE P6.13

6.13.

a) Minimize the states using State Partitioning:

* Output Value:

	S_0	S_1	S_2	S_3	S_4	S_5
AB=00	1	0	1	0	0	0
AB=01	0	1	0	1	1	1
AB=10	1	1	1	1	1	1

101 / Partitioning \ 011

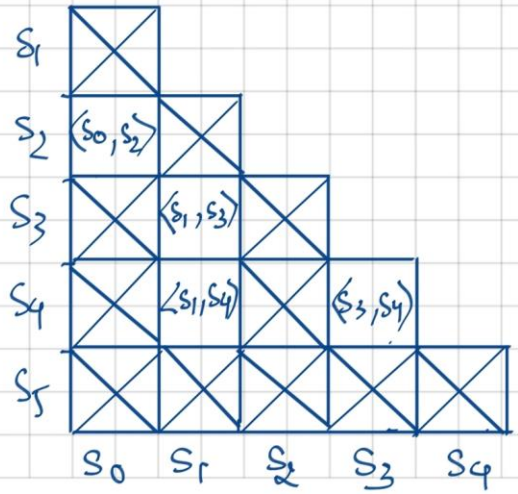
* Next-State:

	S_0	S_1	S_2	S_3	S_4	S_5
AB=00	S_4	S_2	S_1	S_2	S_0	S_2
AB=01	S_2	S_5	S_0	S_5	S_5	S_4
AB=10	S_1	S_4	S_3	S_4	S_1	S_2

	$G_0=\{S_0,S_2\}$	$G_1=\{S_1,S_3,S_4\}$	$G_2=\{S_5\}$
AB=00	G_1G_1	$G_0G_0G_0G_0$	G_0
AB=01	G_0G_0	$G_1G_1G_1G_1$	G_1
AB=10	G_2G_2	$G_1G_1G_1G_0$	G_0

Present State	Next State / Output		
	AB=00	AB=01	AB=10
G_0	$G_1/1$	$G_0/0$	$G_2/1$
G_1	$G_0/0$	$G_1/1$	$G_1/1$
G_2	$G_0/0$	$G_1/1$	$G_0/1$

b) Minimize the states using An Implication Table:



$\Rightarrow$ $\{S_0,S_2\}$

$\{S_1,S_3\}$, $\{S_1,S_4\}$, $\{S_3,S_4\}$

$\{S_5\}$

$\Rightarrow$ $G_0=\{S_0,S_2\}$
$G_1=\{S_1,S_3,S_4\}$
$G_2=\{S_5\}$.

6.19 (Sequential synthesis) Design a recognizer that recognizes an input sequence that has at least three 1's. The recognizer has a single input X and a single output Y, in addition to an asynchronous *Reset* signal. The recognizer sets the output Y to 1 if the input signal X equals 1 at least three clock cycles after *Reset* was disasserted. For the recognizer described above:

(a) Devise the state diagram.

(b) Minimize the number of states.

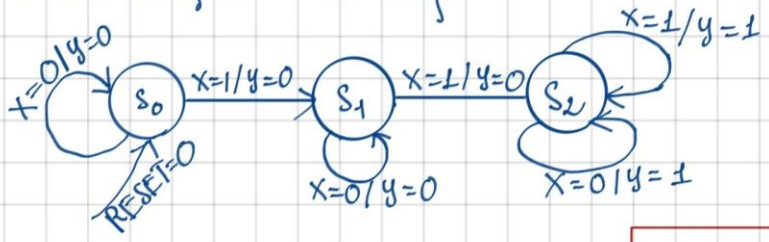
(c) Encode the states to minimize the combinatorial logic.

(d) Draw a schematic diagram using D flip-flops.

6.19



a) State diagram (Mealy)



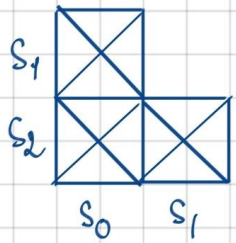
* State Table

Present State	Next State / Output	
	X=0	X=1
S_0	S_0 / 0	S_1 / 0
S_1	S_1 / 0	S_2 / 0
S_2	S_2 / 1	S_2 / 1

* Hỏi có nếu đề y/c 4 bit 1 thì S_0, S_1, S_2, S_3 thì S_1, S_2 có rút gọn được ko?

b) Minimize the number of state

⇒ Không thể rút gọn thêm trạng thái.



c) Encoding state:

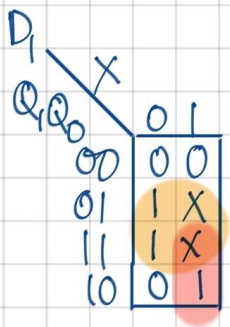
State	Encoding
S_0	00
S_1	01
S_2	10

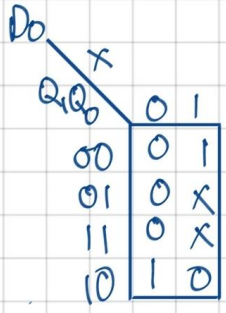
Encoded State Table:

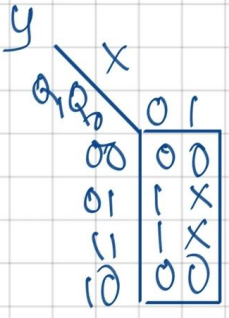
Present State Q_1Q_0	Next state $Q_1^+Q_0^+$ X=0	Next state $Q_1^+Q_0^+$ X=1	Output y
00	00	01	0
01	01	10	0
10	10	10	1

d) D-Flip Flop Excitation Table:

X	Q_1	Q_0	Q_1^+	Q_0^+	D_1	D_0	y
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	1	0	1	0	1
0	1	1	X	X	X	X	X
1	0	0	0	1	0	1	0
1	0	1	1	0	1	0	0
1	1	0	1	0	1	0	1
1	1	1	X	X	X	X	X

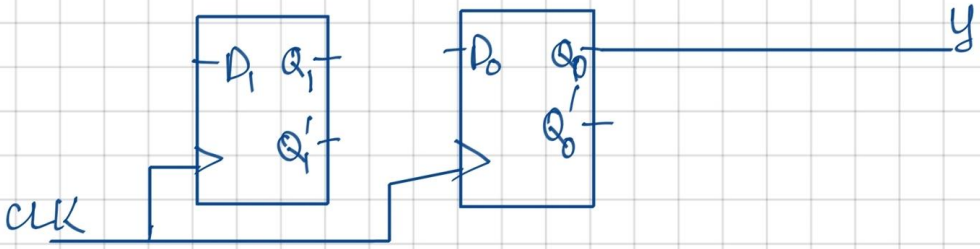


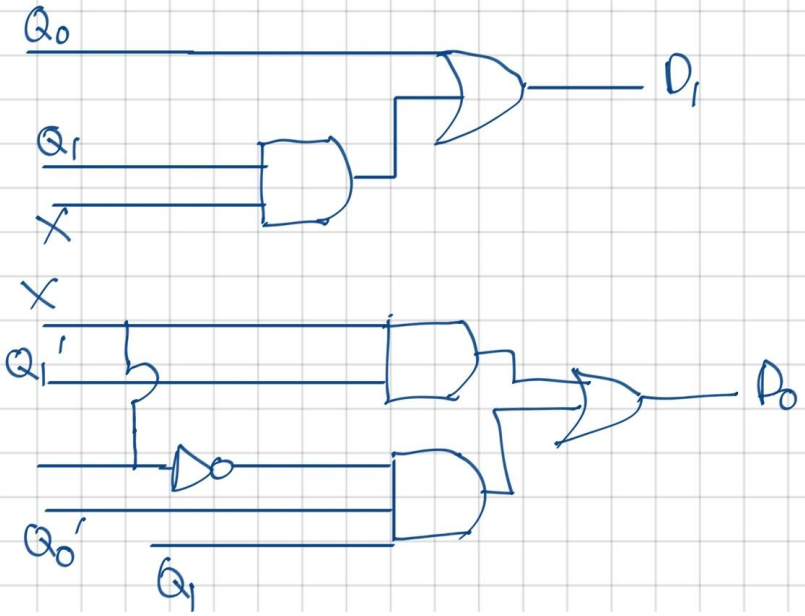




⇒ $D_1 = Q_0 + Q_1X$ ⇒ $D_0 = Q_1'X + Q_1Q_0'X'$ ⇒ $y = Q_0$

* Draw the schematic diagram:

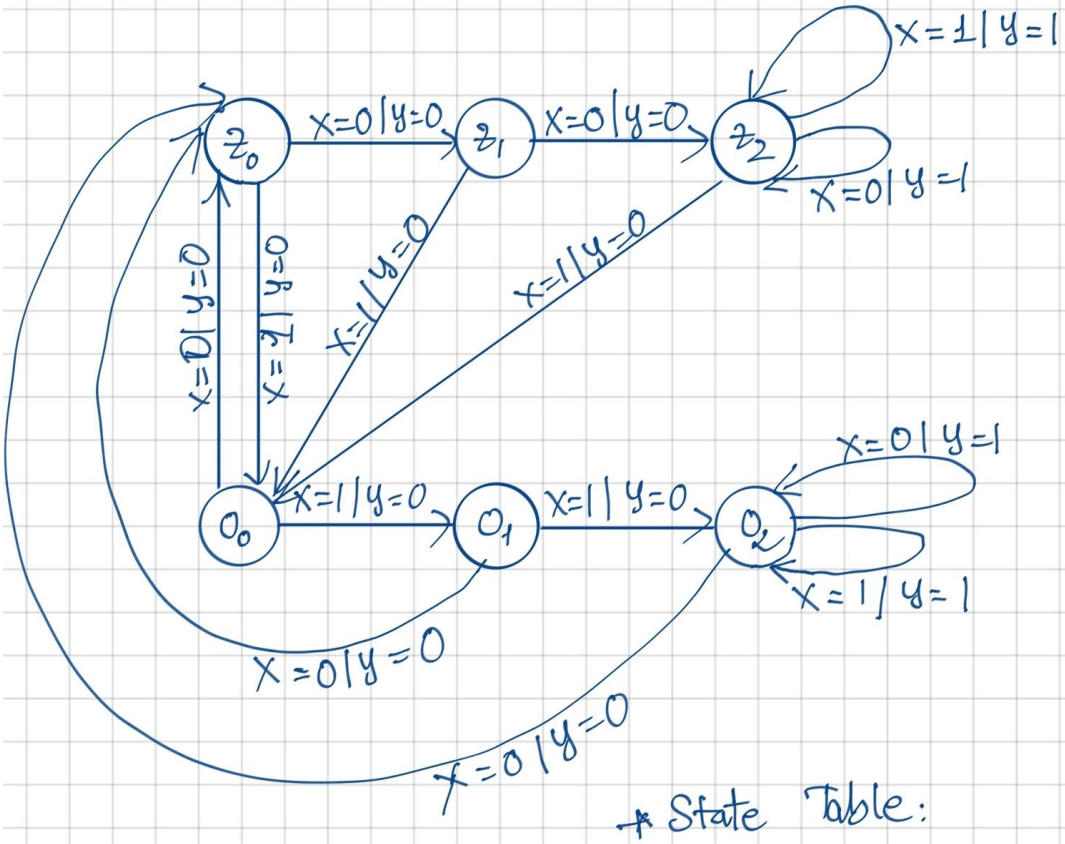




6.20 (Sequential synthesis) Redo Problem 6.19, designing a recognizer that will recognize all the input sequences that have three or more consecutive 1's, or three or more consecutive 0's.

6.20

a) State diagram

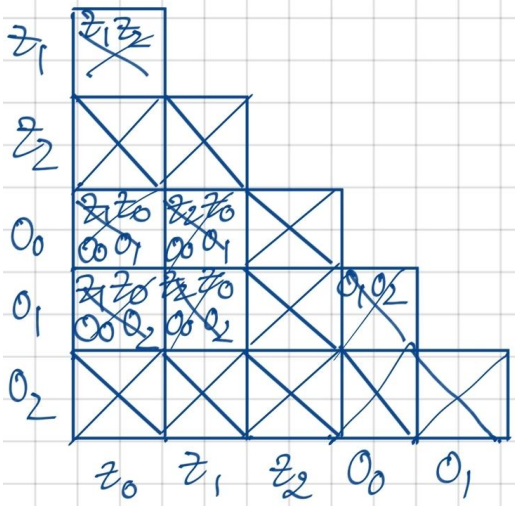


* State Table:

* State table:

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
z_0	z_1	O_0	0	0
z_1	z_2	O_0	0	0
z_2	z_2	O_0	0	0
O_0	z_0	O_1	0	0
O_1	z_0	O_2	0	0
O_2	z_0	O_2	0	1

b)



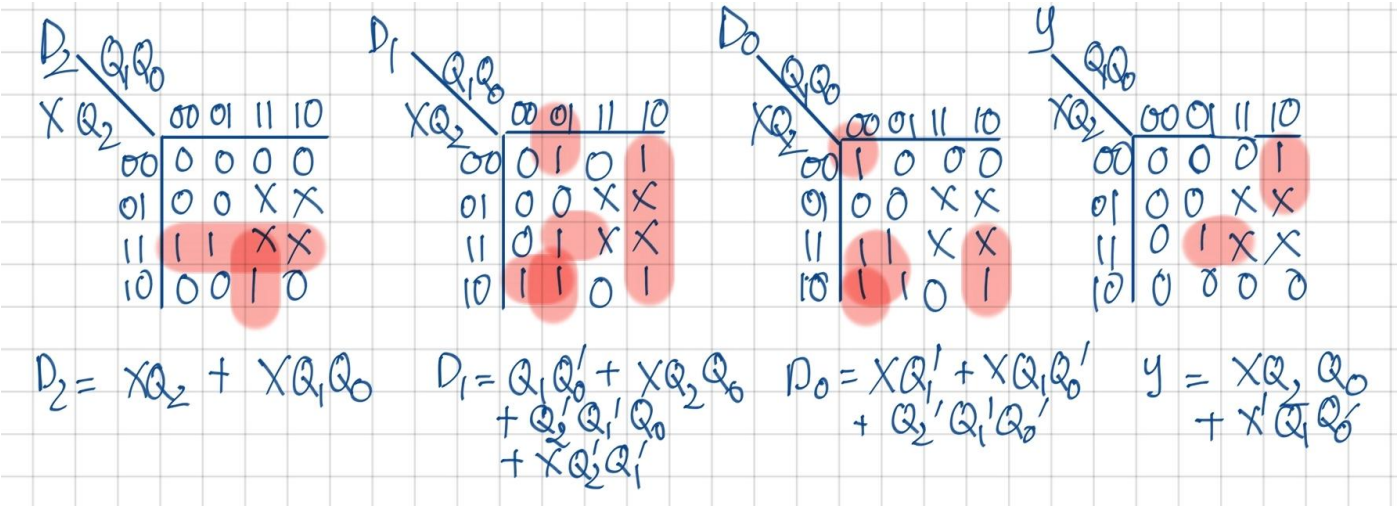
=> Không thể rút gọn thêm trạng thái.

c) Encoding State

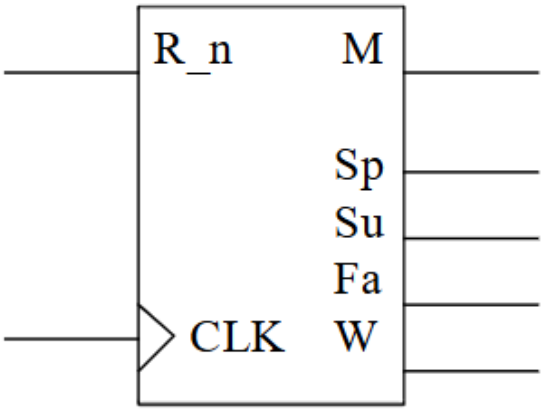
State	Encoding $Q_2Q_1Q_0$
z_0	000
z_1	001
z_2	010
O_0	011
O_1	100
O_2	101

d)

X	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	D_2	D_1	D_0	y
0	0	0	0	0	0	1	0	0	1	0
0	0	0	1	0	1	0	0	1	0	0
0	0	1	0	0	1	0	0	1	0	1
0	0	1	1	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0
0	1	0	1	0	0	0	0	0	0	0
1	0	0	0	0	1	1	0	1	1	0
1	0	0	1	0	1	1	0	1	1	0
1	0	1	0	0	1	1	0	1	1	0
1	0	1	1	1	0	0	1	0	0	0
1	1	0	0	1	0	1	1	0	1	0
1	1	0	1	1	0	1	1	1	1	1



9. Thiết kế một bộ đếm có chức năng đếm các tháng trong một năm Dương Lịch (từ tháng 1 tới tháng 12) và cho biết tháng hiện tại đang ở mùa nào theo sơ đồ khối trong Hình theo Moore FSM.



Hình 2.17: Sơ đồ khối mạch đếm mùa trong năm

Trong đó:

- R_n: Ngõ vào reset đồng bộ tích cực mức thấp, khi R_n tích cực thì bộ đếm sẽ khởi động lại về tháng 1.
- CLK: Ngõ vào clock tích cực cạnh lên của bộ đếm.
- M: Ngõ ra có độ rộng 4 bit thể hiện tháng hiện tại trong năm.
- Sp: Ngõ ra Sp = 1 nghĩa là đang ở mùa xuân.
- Su: Ngõ ra Su = 1 nghĩa là đang ở mùa hè.
- Fa: Ngõ ra Fa = 1 nghĩa là đang ở mùa thu.
- W: Ngõ ra W = 1 nghĩa là đang ở mùa đông.

9.

* State Table

	Next State		Output			
M	R_n=1	R_n=0	Sp	Su	Au	W
S_0	S_1	S_0	1	0	0	0
S_1	S_2	S_0	1	0	0	0
S_2	S_3	S_0	1	0	0	0
S_3	S_4	S_0	0	1	0	0
S_4	S_5	S_0	0	1	0	0
S_5	S_6	S_0	0	1	0	0
S_6	S_7	S_0	0	0	1	0
S_7	S_8	S_0	0	0	1	0
S_8	S_9	S_0	0	0	1	0
S_9	S_{10}	S_0	0	0	0	1
S_{10}	S_{11}	S_0	0	0	0	1
S_{11}	S_0	S_0	0	0	0	1

* Encoded State

M	$Q_3 Q_2$	$Q_1 Q_0$
S_0	0 0	0 1
S_1	0 0	1 0
S_2	0 0	1 1
S_3	0 1	0 0
S_4	0 1	0 1
S_5	0 1	1 0
S_6	0 1	1 1
S_7	1 0	0 0
S_8	1 0	0 1
S_9	1 0	1 0
S_{10}	1 0	1 1
S_{11}	1 1	0 0

* D-FF State Table:

R_n	Q_3	Q_2	Q_1	Q_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+	D_3	D_2	D_1	D_0	Sp	Su	Au	W
1	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0
1	0	0	0	1	0	0	1	0	0	0	1	0	1	0	0	0
1	0	0	1	0	0	0	1	1	0	0	1	1	1	0	0	0
1	0	0	1	1	0	1	0	0	0	1	0	0	1	0	0	0
1	0	1	0	0	0	1	0	1	0	1	0	1	0	1	0	0
1	0	1	0	1	0	1	1	0	0	1	1	0	0	1	0	0
1	0	1	1	0	0	1	1	1	0	1	1	1	0	1	0	0
1	0	1	1	1	1	0	0	0	1	0	0	0	0	0	1	0
1	1	0	0	0	1	0	0	1	1	0	0	1	0	0	1	0
1	1	0	0	1	1	0	1	0	1	0	1	0	0	0	1	0
1	1	0	1	0	1	0	1	1	1	0	1	1	0	0	0	1
1	1	0	1	1	1	1	0	0	1	1	0	0	0	0	0	1
1	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	1
1	1	1	0	1	0	0	0	1	0	0	0	1	0	0	0	0
1	1	1	1	0	0	0	0	1	0	0	0	1	0	0	0	0
1	1	1	1	1	0	0	0	1	0	0	0	1	0	0	0	0

D_3 (rows Q_3Q_2, columns Q_1Q_0):

	$R_n=1$				$R_n=0$			
Q_3Q_2 \ Q_1Q_0	00	01	11	10	00	01	11	10
00	0	0	0	0	0	0	0	0
01	0	0	1	0	0	0	0	0
11	0	0	0	0	0	0	0	0
10	1	1	1	1	0	0	0	0

$\Rightarrow D_3 = R_n \cdot (Q_3Q_2' + Q_3'Q_2Q_1Q_0)$

D_2 (rows Q_3Q_2, columns Q_1Q_0):

	$R_n=1$				$R_n=0$			
Q_3Q_2 \ Q_1Q_0	00	01	11	10	00	01	11	10
00	0	0	1	0	0	0	0	0
01	1	1	0	1	0	0	0	0
11	0	0	0	0	0	0	0	0
10	0	0	1	0	0	0	0	0

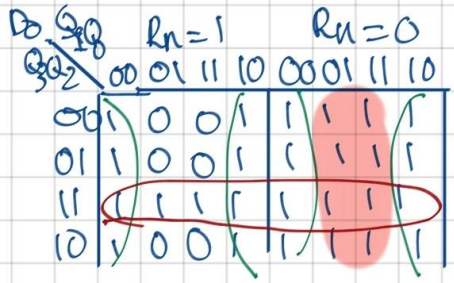
$\Rightarrow D_2 = R_n (Q_3'Q_2Q_1' + Q_3'Q_2Q_0' + Q_2'Q_1Q_0)$

D_1 (rows Q_3Q_2, columns Q_1Q_0):

	$R_n=1$				$R_n=0$			
Q_3Q_2 \ Q_1Q_0	00	01	11	10	00	01	11	10
00	0	1	0	1	0	0	0	0
01	0	1	0	1	0	0	0	0
11	0	0	0	0	0	0	0	0
10	0	1	0	1	0	0	0	0

$\Rightarrow D_1 = R_n (Q_3'Q_1'Q_0 + Q_3'Q_1Q_0' + Q_2'Q_1'Q_0 + Q_2'Q_1Q_0')$

R_n	Q_3	Q_2	Q_1	Q_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+	D_3	D_2	D_1	D_0	Sp. Sufaw
0	0	0	0	0	0	0	0	1	0	0	0	1	0 0 0 0
0	0	0	0	1	0	0	0	1	0	0	0	1	1 0 0 0
0	0	0	1	0	0	0	0	1	0	0	0	1	1 0 0 0
0	0	0	1	1	0	0	0	1	0	0	0	1	1 0 0 0
0	0	1	0	0	0	0	0	1	0	0	0	1	0 1 0 0
0	0	1	0	1	0	0	0	1	0	0	0	1	0 1 0 0
0	0	1	1	0	0	0	0	1	0	0	0	1	0 1 0 0
0	0	1	1	1	0	0	0	1	0	0	0	1	0 0 1 0
0	1	0	0	0	0	0	0	1	0	0	0	1	0 0 1 0
0	1	0	0	1	0	0	0	1	0	0	0	1	0 0 1 0
0	1	0	1	0	0	0	0	1	0	0	0	1	0 0 0 1
0	1	0	1	1	0	0	0	1	0	0	0	1	0 0 0 1
0	1	1	0	0	0	0	0	1	0	0	0	1	0 0 0 1
0	1	1	0	1	0	0	0	1	0	0	0	1	0 0 0 0
0	1	1	1	0	0	0	0	1	0	0	0	1	0 0 0 0
0	1	1	1	1	0	0	0	1	0	0	0	1	0 0 0 0



$\Rightarrow D_0 = Q_0' + Q_1 Q_0 + R_n' Q_0$.